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Nine in Ten, or Four: A Multiverse Analysis of 2,304 Defensible Ways to Compute a Children's Martial-Arts Association's Belt-Progression Rate

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Abstract. Community sport keeps its own records and reports its own performance from them. A metropolitan children's martial-arts association publishes a term progression rate — the share of students who advance a belt in a given term — in each of its annual reports, and its affiliated clubs reproduce that figure in enrolment material. We obtained the association's complete grading register for 18 school terms across 34 clubs (8,911 students, 37,204 grading attempts) and enumerated the ways in which the published quantity can be computed from it. Eight analytic decisions, none of them specified in the association's methodology note, combine into 2,304 defensible specifications. The resulting estimates span 38.6% to 94.1% (median 76.4%, IQR 58.9 to 87.2). The published figure of 91.4% is returned by 74 specifications (3.2%), every one of which takes the students who sat a grading as its denominator; the denominator alone accounts for 58.2% of the variance across specifications. An attempt-level outcome model indicates that tenure carries most of the discrimination (OR 1.31 per additional term enrolled, 95% CI 1.24 to 1.39), while removing the ten technique-criterion scores recorded on the grading sheet changes discrimination by 0.004 AUC (n.s.). Our results suggest that the rate may be more informative when reported alongside the specification that produced it.

Introduction

An association that runs junior sport accumulates a register almost by accident. Attendance is marked because the hall has a capacity, fees are recorded because they must be banked, and grading outcomes are entered because a child's next belt depends on the last one. Once such a register exists, it tends to be asked for a summary figure, and the association studied here supplies one: a term progression rate, published to a single decimal place in each of the four annual reports covering our study window, and quoted without qualification in the enrolment material of most of its clubs.

The quantity is intuitively simple and operationally underdetermined. Whether a student "advanced" depends on whether a stripe awarded mid-term counts; whether a student is in the denominator depends on whether enrolling, being eligible, or actually kneeling on the floor at the grading is the qualifying event; and whether a second attempt after an unsuccessful first counts once or twice depends on a convention the register does not record. The association's published methodology note runs to four sentences and settles none of these questions.

Analytic flexibility of this kind is well documented in research settings [1], [2], and the multiverse approach was developed precisely to make its consequences visible rather than to adjudicate between them [3]. Our contributions are modest and, we suggest, transferable:

- We assemble the association's complete grading register and enumerate the eight undocumented decisions that a computation of its

headline figure must make, yielding 2,304 specifications that we take to be individually defensible.
- We report the distribution of the published quantity across that multiverse and observe that the association's figure, while attainable, sits in the upper tail rather than near the centre.
- We fit an attempt-level outcome model and find, in most settings, that time enrolled rather than the recorded technique criteria carries the model's discrimination.

Background

Multiverse analysis reports an estimate under every reasonable combination of the analytic choices a study leaves open [3], and specification curve analysis formalises the display and inference [4]. Related work quantifies how far estimates move under model choice in observational data [5], [6], and catalogues the decision points at which movement enters [7]. The empirical case is strongest where the same data have been handed to many independent analysts: 29 teams answering one question about disciplinary sanctions reached materially different conclusions [8], and 70 teams analysing a single neuroimaging dataset did likewise [9].

A parallel literature concerns what happens to a measure once it is published. Performance indicators in public services reshape the behaviour they record [10], [11], rankings prove reactive on the organisations they rank [12], and the usefulness of such measures depends heavily on how they are constructed and read [13]. In assessment specifically, a pass rate inherits whatever standard-setting procedure sits behind its cut score [14], [15], rubric-based scoring carries known reliability limits [16], and repeated attempts introduce retest effects that any attempt-selection rule must handle [17]. Prior work at this institution has partitioned assessment outcomes across candidate, examiner, centre and route [18], and has surveyed how far parents trust a displayed inspection tag against how often they check the equipment themselves [19].

Community sport is a plausible setting for both literatures to meet. Dropout from organised youth sport is common and multiply determined [20], [21], and cost is among the barriers parents report [22] — which matters here, because one of the eight decisions concerns students whose fees were waived.

Method

Register. The association provided its grading register for 18 school terms between February 2021 and April 2026, covering 34 affiliated clubs, 8,911 students and 37,204 grading attempts. Each attempt record carries the club, the student identifier, the belt held and sought, ten technique-criterion scores transcribed from the grading sheet, the outcome, the fee status, and the term. Enrolment and attendance were supplied separately and joined on the student identifier. No record was excluded prior to specification.

The multiverse. We enumerated the analytic decisions that a computation of the published quantity must make and that the methodology note leaves open (Table 1). Writing $a_t(s)$ and $d_t(s)$ for the numerator and denominator that specification s produces in term t, the term progression rate is

$$R(s) = \frac{1}{T}\sum_{t=1}^{T}\frac{a_t(s)}{d_t(s)}, \quad s \in \mathcal{S} = \prod_{k=1}^{8} D_k$$

with $|\mathcal{S}| = 2,304$. Every specification was computed on the full register; none was discarded after the fact. We decomposed the variance of $R(s)$ across the eight decision nodes by fitting a linear model of the estimate on the node levels and their two-way interactions.

Decision	Levels	$\lvert D_k\rvert$
Numerator	belts awarded / students advancing / attempts passed	3
Denominator	sat a grading / eligible / enrolled / enrolled, withdrawals retained	4
Attempt selection	any / first / last / best	4
Re-gradings	pooled within term / counted separately	2
Stripe awards	progression / not progression	2
Fee waivers	included / excluded	2
Transfers	retained / dropped	2
Attendance filter	none / 50% / 75%	3

Table 1: The eight decisions a computation of the association's published rate must make, none of which its methodology note specifies. The product of the level counts is 2,304.

Outcome model. To ask what a belt tracks, we fitted a mixed-effects logistic model of the attempt-level outcome with random intercepts for student, club and examiner, and fixed effects for terms enrolled, attendance proportion, fee status, belt sought, and the ten technique-criterion scores. Discrimination was estimated by 1,000 bootstrap replicates at the student level, and the covariate blocks were ablated one at a time.

Results

The 2,304 specifications return term progression rates from 38.6% to 94.1%, with a median of 76.4% and an interquartile range of 58.9 to 87.2. The association's published figure of 91.4% is not an error: 74 specifications (3.2%) return a value at or above it. All 74 take the students who sat a grading as the denominator, which is the choice that separates the distribution most sharply (Figure 1). The variance decomposition attributes 58.2% of the spread to the denominator, 19.7% to attempt selection, and 11.4% to the treatment of withdrawals; the remaining five decisions together account for 6.9%, with 3.8% in two-way interactions.

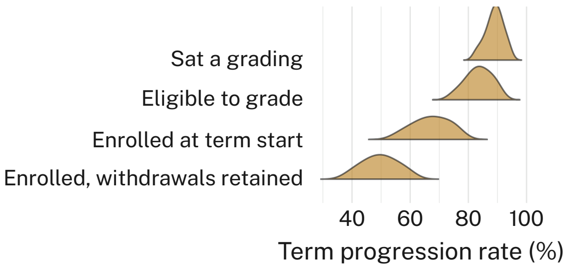

Figure 1: The estimate is set by the denominator before it is set by anything else: no specification taking enrolment at term start returns a figure above 77%, and none taking the students who sat a grading returns one below 82%.

The pattern reproduces within clubs. Every one of the 34 clubs publishes a term progression rate above 85%; under the median specification, none exceeds 81% and eleven fall below 65% (Figure 2). The gap is widest at the clubs with the largest mid-term withdrawal counts, which is the arithmetic one would expect and which no club's published figure reflects.

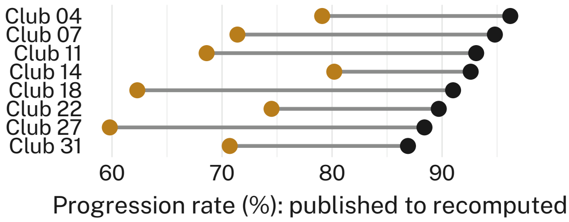

Figure 2: Published club rate (ink) against the median of the 2,304 specifications recomputed from that club's own register (gold), for eight clubs spanning the published range. No recomputed median exceeds 81%.

Attempt selection matters because attempts are not evenly available. A re-grading within the same term costs $48 at 22 of the 34 clubs and nothing at the remaining twelve. Where the fee is waived, 34.6% of students whose first attempt is unsuccessful return for a second within the term; where it is charged, 11.2% do. Under the "best attempt" rule — the rule most consistent with how clubs describe the process to parents — this difference is sufficient to move a club's estimate by up to 7.9 percentage points without any change in what happens on the floor.

The outcome model reaches an AUC of 0.741 (95% CI 0.729 to 0.753). Each additional term enrolled is associated with an odds ratio of 1.31 (95% CI 1.24 to 1.39) for a belt at a given attempt, and attendance proportion with 1.09 per decile (95% CI 1.04 to 1.14). Removing the ten technique-criterion scores degrades discrimination by 0.004 AUC (n.s.); we retain the block for completeness. Removing terms enrolled degrades it by 0.053 (Figure 3).

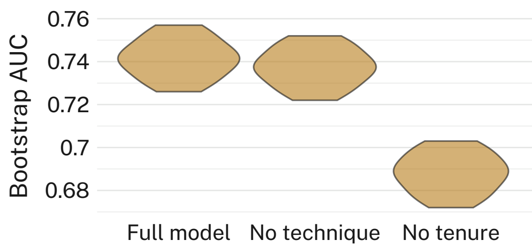

Figure 3: Bootstrap discrimination under two ablations. The technique scores transcribed from the grading sheet can be removed without moving the distribution; terms enrolled cannot.

Discussion

The association's figure is defensible and unrepresentative at the same time, which is the ordinary condition of a statistic computed once under an unrecorded convention. We observe consistent agreement between specifications that share a denominator and consistent divergence between those that do not, and we take this to suggest that publishing the denominator would recover most of the interpretability currently lost. It is worth noting that the register itself is in good order: the spread we report is not a data-quality finding but a reporting-convention one, and the association is better placed than most comparable bodies to adopt a stated specification, having already assembled the material such a specification would operate on. The outcome model's results are consistent with a grading process in which time served is the principal input and the technique criteria are recorded rather than decisive — a reading the criteria's authors may not share, and one our design cannot distinguish from a ceiling effect on the sheet.

Limitations

Our evidence comes from a single association, though the consistency of the pattern across all 34 of its clubs suggests the mechanism is not local to any one of them. The eight decisions we enumerate are those the methodology note leaves open; a fuller note would leave fewer, which would narrow the multiverse without changing the argument for stating one. We do not observe the gradings directly and therefore cannot say whether the technique scores are uninformative or merely uniformly high, although either reading supports reporting them separately from the rate.

Conclusion

A term progression rate computed from one register under 2,304 defensible conventions spans 38.6% to 94.1%. The convention, not the register, is doing most of the work, and the convention is currently unpublished. We suggest that associations reporting such figures state the numerator, the denominator, and the attempt-selection rule alongside the number, and we note that doing so costs nothing that the register does not already contain. Future work might usefully ask what the published figure would have been had the specification been fixed before the first term rather than after the fourth annual report.

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